

MULTIVARIABLE CALCULUS

ASSIGNMENT - 1

Muhammad Zain Naved.

2022F - BCS - 160

Section - C.

14.7 (Extreme Values)

$$f(x, y) = x^2 + xy + y^2 + 3x - 3y + 4$$

$$f_x = 2x + y + 3, \quad f_y = x + 2y - 3, \quad f_{xy} = 1$$
$$f_{xx} = 2, \quad f_{yy} = 2$$

For critical point,

$$f_x = 0 \quad 2x + y + 3 = 0 \quad \text{--- (i)}$$

$$f_y = 0 \quad x + 2y - 3 = 0 \quad \text{--- (ii)}$$

multiply (i) by 2 & subtract by (ii)

$$\begin{array}{r} 4x + 2y + 6 = 0 \\ -x + 2y - 3 = 0 \\ \hline \end{array}$$

$$3x + 9 = 0$$

$$x = \frac{-9}{3} = -3.$$

$$2(-3) + y + 3 = 0$$

$$-6 + 3 + y = 0$$

$$y - 3 = 0 \Rightarrow y = 3, \quad x = -3$$

Now,

$$f_{xx}f_{yy} - (f_{xy})^2 = 2(2) - 1^2 = 4 - 1 = 3 > 0.$$

minima.

$$f(-3, 3) = 9 - 9 + 9 - 9 - 9 + 4$$
$$= -5.$$



Teacher's Signature

$$f(x, y) = x^2 - 4xy + y^2 + 6y + 2.$$

$$f_x = 2x - 4y$$

$$f_{xx} = 2$$

$$f_y = -4x + 2y + 6$$

$$f_{yy} = 2$$

$$f_{xy} = -4$$

For critical point,

$$f_x = 2x - 4y = 0 \quad \text{--- (i)}$$

$$f_y = -4x + 2y + 6 = 0 \quad \text{--- (ii)}$$

multiplying (i) by 2 and adding (ii).

$$-4x + 2y + 6 = 0$$

$$4x - 4y = 0$$

$$-2y + 6 = 0$$

$$y = \frac{6}{2} = 3.$$

$$2x - 4(3) = 0$$

$$2x - 12 = 0$$

$$x = \frac{12}{2} = 6.$$

(6, 3) are the points.

$$f_{xx}f_{yy} - (f_{xy})^2 = 2(2) - (-4)^2$$

$$= 4 - 16 = -12 < 0$$

Saddle point.

14.8 (Lagrange Multiplier)

Extrema on the ellipse. Find the points on the ellipse $x^2 + 2y^2 = 1$ where $f(x, y) = xy$ has its extreme values.

$$f(x, y) = xy$$

$$g(x, y) = x^2 + 2y^2 - 1 = 0$$

we have to find the extreme value of $f(x, y)$ subject to constraint $g(x, y)$.

$$\nabla f = \lambda \nabla g.$$

$$\nabla f = \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j}$$

$$\nabla f = y \hat{i} + x \hat{j}$$

Now,

$$\begin{aligned} \nabla g &= \frac{\partial g}{\partial x} \hat{i} + \frac{\partial g}{\partial y} \hat{j} \\ &= 2x \hat{i} + 4y \hat{j} \end{aligned}$$

So,

$$\lambda(\nabla g) = 2\lambda x \hat{i} + 4\lambda y \hat{j}$$

$$\nabla f = \lambda \nabla g.$$

$$y \hat{i} + x \hat{j} = (2\lambda x) \hat{i} + (4\lambda y) \hat{j}$$

Comparing 'i' and 'j' components.

$$y = 2\lambda x$$

$$x = 4\lambda y$$

put value of x in y .

$$y = 2\lambda(4\lambda y) \Rightarrow 8\lambda^2 y$$

$$y = 8\lambda^2 y \Rightarrow \lambda^2 = \frac{1}{8}$$

$$\lambda = \pm \frac{1}{2\sqrt{2}}$$

$$\Rightarrow x = \left(\pm \frac{1}{2\sqrt{2}} \right) y = \pm \frac{1}{\sqrt{2}} y$$

$$g(x, y) = x^2 + 2y^2 - 1 = 0$$

$$= \left(\pm \frac{1}{\sqrt{2}} \right)^2 y^2 + 2y^2 - 1 = 0$$

$$= \left(\pm \frac{\sqrt{2}\sqrt{2}}{\sqrt{2}} \right) y^2 + 2y^2 - 1 = 0$$

$$= \left(\pm \sqrt{2} \right) y^2 + 2y^2 - 1 = 0$$

$$= \pm 2y^2 + 2y^2 - 1 = 0$$

$$= \pm 4y^2 - 1 = 0 \Rightarrow y^2 = \pm \frac{1}{4}$$

$$y = \pm \frac{1}{2}$$

So,

$$x = \pm \frac{1}{\sqrt{2}} \left(\pm \frac{1}{2} \right)$$

$$x = \pm \frac{\sqrt{2}}{2}$$

Therefore, f takes on its extreme values at $\left(\pm \frac{\sqrt{2}}{2}, \frac{1}{2} \right)$ and $\left(\pm \frac{\sqrt{2}}{2}, -\frac{1}{2} \right)$. The extreme values of f on the ellipse are $\pm \frac{\sqrt{2}}{2}$.

Extrema on a circle. Find the extreme values of $f(x, y) = xy$ subject to the constraint $g(x, y) = x^2 + y^2 - 10 = 0$.

$$f(x, y) = xy.$$

$$g(x, y) = x^2 + y^2 - 10 = 0.$$

$$\nabla f = \lambda \nabla g.$$

$$\frac{\partial f}{\partial x} i + \frac{\partial f}{\partial y} j = \lambda \left\{ \frac{\partial g}{\partial x} i + \frac{\partial g}{\partial y} j \right\}$$

$$y i + x j = \lambda [2x i + 2y j]$$

Comparing.

$$y = 2\lambda x$$

$$x = 2\lambda y$$

$$y = 2\lambda (2\lambda y)$$

$$\Rightarrow y = 4\lambda^2 y$$

$$\lambda^2 = \frac{1}{4}$$

$$\Rightarrow \lambda = \pm \frac{1}{2}$$

put " λ " in " x "

$$x = 2y \left(\pm \frac{1}{2} \right)$$

$$\Rightarrow x = \pm y$$

$$g(x, y) = x^2 + y^2 - 10 = 0$$

$$\text{as } x = \pm y.$$

$$= y^2 + y^2 - 10 = 0$$

$$= 2y^2 = 10$$

$$\Rightarrow y = \pm \sqrt{5}$$

$$\text{as } x = \pm \sqrt{5}$$

The value of x, y and λ .

Therefore f takes on its extreme values at $(\pm \sqrt{5}, \sqrt{5})$, $(\pm \sqrt{5}, -\sqrt{5})$. The extreme values of f on the circle are $\pm \sqrt{5}$.

15.5 (Triple Integrals)

$$\int_0^1 \int_0^1 \int_0^1 (x^2 + y^2 + z^2) dz dy dx$$

$$\int_0^1 \int_0^1 \left[\int_0^1 x^2 dz + \int_0^1 y^2 dz + \int_0^1 z^2 dz \right] dy dx$$

$$\int_0^1 \int_0^1 \left[x^2 \int_0^1 dz + y^2 \int_0^1 dz + \int_0^1 z^2 dz \right] dy dx$$

$$\int_0^1 \int_0^1 \left[x^2 (z) \Big|_0^1 + y^2 (z) \Big|_0^1 + \frac{z^3}{3} \Big|_0^1 \right] dy dx$$

$$\int_0^1 \int_0^1 \left[x^2 (1-0) + y^2 (1-0) + \frac{1}{3} \right] dy dx$$

$$\int_0^1 \int_0^1 (x^2 + y^2 + \frac{1}{3}) dy dx$$

$$\int_0^1 \left[\int_0^1 x^2 dy + \int_0^1 y^2 dy + \int_0^1 \frac{1}{3} dy \right] dx$$

$$\int_0^1 \left[x^2 \int_0^1 dy + \int_0^1 y^2 dy + \frac{1}{3} \int_0^1 dy \right] dx$$

$$\int_0^1 \left[x^2 (y) \Big|_0^1 + \frac{y^3}{3} \Big|_0^1 + \frac{1}{3} y \Big|_0^1 \right] dx$$

$$\int_0^1 \left[x^2 (1-0) + \frac{1}{3} + \frac{1}{3} (1-0) \right] dx$$

$$\int_0^1 x^2 + \frac{1}{3} + \frac{1}{3} dx$$

$$\int_0^1 x^2 dx + \int_0^1 \frac{2}{3} dx$$

$$\frac{x^3}{3} \Big|_0^1 + \frac{2}{3} (x) \Big|_0^1$$

$$\frac{1}{3} + \frac{2}{3} = \frac{3}{3} = 1$$

$$\int_0^{\sqrt{2}} \int_0^{3y} \int_{x^2+3y^2}^{8-x^2-y^2} dz dx dy.$$

$$= \int_0^{\sqrt{2}} \int_0^{3y} z \Big|_{x^2+3y^2}^{8-x^2-y^2} dx dy$$

$$= \int_0^{\sqrt{2}} \int_0^{3y} (8-x^2-y^2-x^2-3y^2) dx dy.$$

$$= \int_0^{\sqrt{2}} \int_0^{3y} (8-2x^2-4y^2) dx dy.$$

$$= \int_0^{\sqrt{2}} \left[8 \int_0^{3y} dx - 2 \int_0^{3y} x^2 dx - 4 \int_0^{3y} y^2 dx \right] dy$$

$$= \int_0^{\sqrt{2}} \left[8(x) \Big|_0^{3y} - 2 \frac{x^3}{3} \Big|_0^{3y} - 4 y^2 x \Big|_0^{3y} \right] dy$$

$$= \int_0^{\sqrt{2}} \left[8(3y-0) - \frac{2}{3} (3y)^3 - 4y^2 (3y) \right] dy$$

$$= \int_0^{\sqrt{2}} \left[24y - \frac{2}{3} (27y^3) - 12y^3 \right] dy$$

$$= \int_0^{\sqrt{2}} 24y - 18y^3 - 12y^3 dy$$

$$= \int_0^{\sqrt{2}} 24y - 30y^3 dy$$

$$= 24 \int_0^{\sqrt{2}} y dy - 30 \int_0^{\sqrt{2}} y^3 dy$$

$$= 24 \frac{y^2}{2} \Big|_0^{\sqrt{2}} - 30 \frac{y^4}{4} \Big|_0^{\sqrt{2}}$$

$$= \frac{24}{2} (\sqrt{2})^2 - \frac{30}{4} (\sqrt{2})^4$$

$$= 12(2) - \frac{30}{4} (\sqrt{2})^4$$

$$= 24 - \frac{30}{4} (4)$$

$$= -6.$$

16.4 (Green's Theorem)

Find the K -component of $\text{curl}(F)$ for vector fields on the plane.

$$F = (u+y)i + (2xy)j$$

$$M = (u+y)i + (2xy)j$$

$M \qquad N$

$$K = \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}$$

$$\text{Curl}(F) = 2y - 1.$$

$$F = (ue^x)i + (ye^x)j$$

$$\text{curl}(F) = \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}$$

$$\text{curl}(F) = ye^x - ue^x$$

Example - 3.

Verify both forms of Green's Theorem for the vector field.

$$F(u,y) = (u-y)i + xj$$

and region R bounded by the unit circle

$$C: r(t) = (\cos t)i + (\sin t)j, \quad 0 \leq t \leq 2\pi.$$

First evaluate the counter clockwise circulation of

$$F = Mi + Nj \text{ around } C.$$

$$M = u - y = \cos t - \sin t$$

$$N = u = \cos t$$

$$dx = d(\cos t) = -\sin t dt$$

$$dy = d(\sin t) = \cos t dt.$$

Therefore,

$$\oint_C F \cdot dr = \oint_C M dx + N dy.$$

$$= \int_0^{2\pi} (\cos t - \sin t) dt (-\sin t) + (\cos t)(\cos t) dt$$

$$= \int_0^{2\pi} (-\sin t \cos t + 1) dt = 2\pi$$

Next, we find the curl integral,

Since, $M = x - y$, and $N = x$, we have

$$\frac{\partial M}{\partial x} = 1, \quad \frac{\partial M}{\partial y} = -1, \quad \frac{\partial N}{\partial x} = 1, \quad \frac{\partial N}{\partial y} = 0.$$

Therefore,

$$\begin{aligned} \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy &= \iint_R (1 - (-1)) dx dy \\ &= 2 \iint_R dx dy = 2 (\text{area inside the unit circle}) = 2\pi \end{aligned}$$

Now, starting with the outward flux.

$$\begin{aligned} \oint_C M dy - N dx &= \int_0^{2\pi} (\cos t - \sin t)(\cos t dt) - (\cos t)(-\sin t dt) \\ &= \int_0^{2\pi} \cos^2 t dt = \pi \end{aligned}$$

Next, we compute divergence integral:

$$\iint_R \left(\frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} \right) dx dy = \iint_R (1+0) dx dy = \iint_R dx dy = \pi.$$

Hence, both the R.H.S and L.H.S equal, as asserted by the flux divergence version of Green's Theorem.

16.2 (Vector Integrals, Line Integrals).

Example - 3.

Evaluate the line integral $\int_C -y dx + z dy + 2x dz$, where C is the helix $r(t) = (\cos t)i + (\sin t)j + tK$, $0 \leq t \leq 2\pi$.

We express everything in terms of the parameter t , so $x = \cos t$, $y = \sin t$, $z = t$, and $dx = -\sin t dt$, $dy = \cos t dt$, $dz = dt$. Then,

$$\begin{aligned}\int_C -y dx + z dy + 2x dz &= \int_0^{2\pi} [(-\sin t)(-\sin t) + t \cos t + 2 \cos t] dt \\&= \int_0^{2\pi} [2 \cos t + t \cos t + \sin^2 t] dt \\&= \left[2 \sin t + (t \sin t + \cos t) + \left(\frac{t}{2} - \frac{\sin 2t}{4} \right) \right]_0^{2\pi} \\&= [0 + (0 + 1) + (\pi - 0)] - [0 + (0 + 1) + (0 - 0)] \\&= \pi.\end{aligned}$$

Example - 4.

Find the work done by the force field $F = (y - x^2)i + (z - y^2)j + (x - z^2)k$ in moving an object along the curve $r(t) = ti + t^2j + t^3k$, $0 \leq t \leq 1$, from $(0, 0, 0)$ to $(1, 1, 1)$.

First we evaluate F on the curve $r(t)$:

$$\begin{aligned}F &= (y - x^2)i + (z - y^2)j + (x - z^2)k \\&= (t^2 - t^2)i + (t^3 - t^4)j + (t - t^6)k.\end{aligned}$$

Then we find $\frac{dr}{dt}$,

$$\frac{dr}{dt} = \frac{d}{dt} (ti + t^2j + t^3k) = i + 2tj + 3t^2k.$$

Finally, we find $F \cdot dr/dt$ and integrate from $t=0$ to $t=1$.

$$F \cdot \frac{dr}{dt} = [(t^3 - t^4) + (t - t^6)k] \cdot (i + 2tj + 3t^2k)$$

$$= (t^3 - t^4)(2t) + (t - t^6)(3t^2) = 2t^4 - 2t^5 + 3t^3 - 3t^8$$

So,

$$\text{Work} = \int_a^b F \cdot \frac{dr}{dt} \cdot dt = \int_0^1 (2t^4 - 2t^5 + 3t^3 - 3t^8) dt$$

$$= \left[\frac{2}{5} t^5 - \frac{2}{6} t^6 + \frac{3}{4} t^4 - \frac{3}{9} t^9 \right]_0^1 = \frac{29}{60}$$

Example - 5

Find the work done by the force field $F = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ in moving an object along the curve C parametrized by $r(t) = \cos(\pi t)\mathbf{i} + t^2\mathbf{j} + \sin(\pi t)\mathbf{k}$, $0 \leq t \leq 1$.

$$F(r(t)) = \cos(\pi t)\mathbf{i} + t^2\mathbf{j} + \sin(\pi t)\mathbf{k}$$

Next, we compute $\frac{dr}{dt}$

$$\frac{dr}{dt} = -\pi \sin(\pi t)\mathbf{i} + 2t\mathbf{j} + \pi \cos(\pi t)\mathbf{k}$$

we then calculate the dot product.

$$F(r(t)) \cdot \frac{dr}{dt} = -\pi \sin(\pi t) \cos(\pi t) + 2t^3 + \pi \sin(\pi t) \cos(\pi t)$$

$$= 2t^3$$

The work done in the line integral

$$\int_a^b F(r(t)) \cdot \frac{dr}{dt} dt = \int_0^1 2t^3 dt = \left[\frac{t^4}{2} \right]_0^1 = \frac{1}{2}$$

Example - 6.
 at fluid's velocity is $F = xi + zj + yk$. Find the flow
 along the helix $r(t) = (\cos t)i + (\sin t)j + tK$, $0 \leq t \leq \pi/2$

We evaluate F on the curve $r(t)$

$$F = xi + zj + yk = (\cos t)i + tj + (\sin t)k$$

and then find $\frac{dr}{dt}$

$$\frac{dr}{dt} = (-\sin t)i + (\cos t)j + k$$

The dot product of F with $\frac{dr}{dt}$ is

$$\begin{aligned} F \cdot \frac{dr}{dt} &= \cos t(-\sin t) + (t)(\cos t) + (\sin t)1 \\ &= -\sin t \cos t + t \cos t + \sin t. \end{aligned}$$

Finally, we integrate $F \cdot \frac{dr}{dt}$ from $t=0$, to $t = \frac{\pi}{2}$.

$$\text{Flow} = \int_a^b F \cdot \frac{dr}{dt} dt = \int_0^{\pi/2} (-\sin t \cos t + t \cos t + \sin t) dt$$

$$= \left[\frac{\cos^2 t}{2} + t \sin^2 t \right]_0^{\pi/2} = \left(0 + \frac{\pi}{2} \right) - \left(\frac{1}{2} + 0 \right)$$

$$= \frac{\pi}{2} - \frac{1}{2}$$

Example - 7

Find the circulation of the field $F = (x-y)i + xj$ around
 the circle $r(t) = (\cos t)i + (\sin t)j$, $0 \leq t \leq 2\pi$.

$$F = (x-y)i + xj = (\cos t - \sin t)i + (\cos t)j, \text{ and}$$

$$\frac{dr}{dt} = (-\sin t)i + (\cos t)j$$

$$F \cdot \frac{dr}{dt} = -\sin t \cos t + \sin^2 t + \cos^2 t$$

NOTE

$$\begin{aligned} \text{Circulation} &= \int_0^{2\pi} \frac{F \cdot dr}{dt} dt = \int_0^{2\pi} (1 - \sin t \cos t) dt \\ &= \left[t - \frac{\sin^2 t}{2} \right]_0^{2\pi} = 2\pi. \end{aligned}$$

ASSIGNMENT - 2

Page : 999. 16.5 (surfaces and Areas)

Example - 1

Find a parameterization of the cone

$$z = \sqrt{x^2 + y^2}, \quad 0 \leq z \leq 1$$

$$x = r \cos \theta,$$

$$y = r \sin \theta,$$

$$z = \sqrt{x^2 + y^2} = r, \quad \text{with } 0 \leq r \leq 1 \text{ \& } 0 \leq \theta \leq 2\pi.$$

Taking $u = r$ \& $v = \theta$

$$r(r, \theta) = (r \cos \theta)i + (r \sin \theta)j + r k$$

$$0 \leq r \leq 1, \quad 0 \leq \theta \leq 2\pi$$

The parameterization is one-to-one on the interior of the domain R , though not on the boundary tip of its cone where $r = 0$.

Example - 2.

Find a parameterization of the sphere

$$x^2 + y^2 + z^2 = a^2.$$

$$x = a \sin \phi \cos \theta,$$

$$y = a \sin \phi \sin \theta,$$

$$z = a \cos \phi, \quad 0 \leq \phi \leq \pi, \quad 0 \leq \theta \leq 2\pi.$$

Taking $u = \phi$ \& $v = \theta$

$$r(\phi, \theta) = (a \sin \phi \cos \theta)i + (a \sin \phi \sin \theta)j + (a \cos \phi)k$$

$$0 \leq \phi \leq \pi, \quad 0 \leq \theta \leq 2\pi$$

Parameterization is one-to-one, though not on its boundary "pole" where $\phi = 0$ or $\phi = \pi$.

Example - 3.

Find the parametrization of the cylinder.

$$x^2 + (y-3)^2 = 9, \quad 0 \leq z \leq 5$$

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$z = z$$

For the given cylinder.

$$x^2 + (y-3)^2 = 9$$

$$r^2 - 6r \sin \theta + 9 = 0.$$

A typical point on the cylinder therefore has,

$$x = r \cos \theta = 6 \sin \theta \cos \theta = 3 \sin 2\theta$$

$$y = r \sin \theta = 6 \sin^2 \theta$$

$$z = z.$$

Taking $u = \theta$ & $v = z$ in eq. gives the one-to-one parametrization.

$$r(\theta, z) = (3 \sin 2\theta) i + (6 \sin^2 \theta) j + z k.$$

$$0 \leq \theta \leq \pi, \quad 0 \leq z \leq 5.$$

Solved Example from ppt (Stoke's Theorem)

Use Stoke's Theorem to calculate the circulation of the field $F = x^2 i + 2xy j + z^2 k$ around the curve C . The ellipse $4x^2 + y^2 = 4$ in the xy plane counter clockwise when viewed from above.

The surface S is $4x^2 + y^2 = 4$ i.e. $f = 4x^2 + y^2 = 4$.

$$\nabla \times F = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 & 2xy & z^2 \end{vmatrix}$$

$$(\nabla \times F) \cdot K = 2$$

$$\text{Circulation } \oint_C F \cdot dr = \int_0^2 \int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} 2 dx dy = 2 \int_0^2 \sqrt{4-y^2} dy.$$

$$= 2 \left(\frac{1}{2} \sqrt{4-y^2} + \frac{4}{2} \frac{\sin^{-1} y}{2} \right)^2$$

$$= 4 \left(\frac{x}{2} - 0 \right) = 2x$$

16.8 (Divergence Theorem)

Example - 3.

Find the flux of $F = xy\mathbf{i} + yz\mathbf{j} + xz\mathbf{k}$ outward through the surface of the cube cut from the first octant by the planes $x=1$, $y=1$, and $z=1$.

$$\nabla \cdot F = \frac{\partial}{\partial x}(xy) + \frac{\partial}{\partial y}(yz) + \frac{\partial}{\partial z}(xz) = y + z + x$$

over the cube's interior,

$$\text{Flux} = \iint_{\substack{\text{cube} \\ \text{surface}}} F \cdot \mathbf{n} \, d\sigma = \iiint_{\substack{\text{cube} \\ \text{interior}}} \nabla \cdot F \, dV$$

$$= \int_0^1 \int_0^1 \int_0^1 (x+y+z) \, dx \, dy \, dz = \frac{3}{2}$$

14.6 (Tangent Planes)

$$x^2 + y^2 + z^2 = 3, \quad P_0(1, 1, 1).$$

$$f_x = 2x \quad \text{at} \quad (1, 1, 1) = 2(1) = 2.$$

$$f_y = 2y \quad \text{at} \quad (1, 1, 1) = 2(1) = 2.$$

$$f_z = 2z \quad \text{at} \quad (1, 1, 1) = 2(1) = 2.$$

Tangent Plane.

$$f_x(P_0)(x-x_0) + f_y(P_0)(y-y_0) + f_z(P_0)(z-z_0) = 0.$$

$$2(x-1) + 2(y-1) + 2(z-1) = 0$$

$$2x - 2 + 2y - 2 + 2z - 2 = 0$$

$$2x + 2y + 2z - 6 = 0.$$

Normal Line.

$$x = 1 + 2t$$

$$y = 1 + 2t$$

$$z = 1 + 2t$$

$$x = x_0 + f_x(p_0)t$$

$$y = y_0 + f_y(p_0)t$$

$$z = z_0 + f_z(p_0)t$$

$$x^2 + y^2 - z^2 = 18, \quad p_0(3, 5, -4).$$

$$f_x = 2x \text{ at } (3, 5, -4) = 2(3) = 6$$

$$f_y = 2y \text{ at } (3, 5, -4) = 2(5) = 10$$

$$f_z = -2z \text{ at } (3, 5, -4) = 2(-4) = -8.$$

Tangent Plane.

$$f_x(p_0)(x-x_0) + f_y(p_0)(y-y_0) + f_z(p_0)(z-z_0) = 0$$

$$6(x-3) + 10(y-5) + (-8)(z+4) = 0$$

$$6x - 18 + 10y - 50 - 8z - 32 = 0$$

$$6x + 10y - 8z - 100 = 0.$$

Normal Line.

$$x = 3 + 8t$$

$$y = 5 + 10t$$

$$z = -4 - 8t.$$

Partial Derivatives. (14.4)

$$w = x^2 + y^2, \quad x = \cos t, \quad y = \sin t, \quad t = \pi.$$

$$\frac{dw}{dt} = \frac{dw}{dx} \cdot \frac{dx}{dt} + \frac{dw}{dy} \cdot \frac{dy}{dt}$$

$$= 2x \cdot (-\sin t) + 2y \cdot (\cos t)$$

$$= -2x \sin t + 2y \cos t$$

For applying limit, put values of x, y .

$$\frac{dw}{dt} = -2 \cos t \sin t + 2 \sin t \cos t$$

$$= 0.$$

$$\left(\frac{dw}{dt} \right)_{\pi} = 0.$$

$$w = x^2 + y^2, \quad x = \cos t + \sin t, \quad y = \cos t - \sin t; \quad t = 0.$$

$$\begin{aligned} \frac{dw}{dt} &= \frac{\partial w}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial w}{\partial y} \cdot \frac{dy}{dt} \\ &= 2x (-\sin t + \cos t) + 2y (-\sin t - \cos t) \end{aligned}$$

putting the values,

$$\begin{aligned} &= 2 \cos t + 2 \sin t (-0 + 1) + 2 \cos t - 2 \sin t (-0 - 1) \\ &= 2 \cos(0) + 2 \cos(1) (-1) \\ &= 2(1) - 2(1) \\ &= 0. \end{aligned}$$

ASSIGNMENT - 3

6.1 (Laplace Transform).

Example - 1

Let $f(t) = 1$ when $t \geq 0$, Find $F(s)$.

$$f(t) = \mathcal{L}^{-1}(F)$$

$$F(s) = \int_0^{\infty} K(s, t), f(t) dt.$$

$$\therefore K(s, t) = e^{-st}$$

So,

$$\mathcal{L}(f) = \mathcal{L}(1) = \int_0^{\infty} e^{-st} dt = -\frac{1}{s} e^{-st} \Big|_0^{\infty} \Rightarrow \frac{1}{s}$$

Such integral is called

($s > 0$)

improper integral.

$$\int_0^{\infty} e^{-st} f(t) dt = \lim_{T \rightarrow \infty} \int_0^T e^{-st} f(t) dt.$$

Hence,

$$\begin{aligned} \int_0^{\infty} e^{-st} dt &= \lim_{T \rightarrow \infty} \left[-\frac{1}{s} e^{-st} \right]_0^T \\ &= \lim_{T \rightarrow \infty} \left[-\frac{1}{s} e^{-sT} + \frac{1}{s} e^0 \right] \Rightarrow \frac{1}{s} \end{aligned}$$

Example - 2.

Laplace Transformation $\mathcal{L}(e^{at})$ of the Exponential f.

Let $f(t) = e^{at}$, when $t \geq 0$, where 'a' is constant. Find $\mathcal{L}(f)$.

$$\mathcal{L}(e^{at}) = \int_0^{\infty} e^{-st} e^{at} dt = \frac{1}{a-s} e^{-(s-a)t} \Big|_0^{\infty};$$

hence, when $s-a > 0$

$$\mathcal{L}(e^{at}) = \frac{1}{s-a}$$

6.3

2. t ($0 < t < 2$)

In terms of unit step function.

$$f(t) = t(u(t) - u(t-2))$$

Transform of the function.

$$\begin{aligned}\mathcal{L}\{f(t)\} &= \mathcal{L}\{t(u(t)) - u(t-2)\} \\ &= \mathcal{L}\{tu(t)\} - \mathcal{L}\{t(u)(t-2)\} \\ &= \frac{1}{s^2} - e^{-2s} \left(\frac{1}{s^2} - \frac{2}{s} \right) \\ &= \frac{1}{s^2} - \frac{e^{-2s}}{s^2} + \frac{2e^{-2s}}{s} \\ &= \frac{1 - e^{-2s} + 2se^{-2s}}{s^2}\end{aligned}$$

3- $t - 2 \ (t > 2)$

In terms of unit step function.

$$f(t) = (t-2) u(t-2)$$

Transform of the function.

$$\mathcal{L}\{f(t)\} = \mathcal{L}\{(t-2) u(t-2)\}$$

$$\therefore \mathcal{L}\{f(t-a) u(t-a)\} = e^{-as} \mathcal{L}\{f(t)\}.$$

So,

$$= \frac{e^{-2s}}{s^2}$$

4- $\cos 4t \ (0 < t < \pi)$.

In terms of unit step function.

$$f(t) = \cos 4t (u(t) - u(t-\pi))$$

Transform of the function.

$$\mathcal{L}\{f(t)\} = \mathcal{L}\{\cos 2t u(t)\} - \mathcal{L}\{\cos 2t u(t-\pi)\}$$

$$= \frac{s}{s^2 - 4} - e^{-\pi s} \left(\frac{s}{s^2 - 4} \right)$$

$$= \frac{s}{s^2 - 4} (1 - e^{-\pi s})$$

Date _____

11.9 (page 513 - 514 Example) [from Fourier series to Fourier integral]

Find the Fourier integral representation of the function,

$$f(x) = \begin{cases} 1 & \text{if } |x| < 1 \\ 0 & \text{if } |x| > 1. \end{cases}$$

$$A(\omega) = \frac{1}{\pi} \int_{-\infty}^{\infty} f(v) \cos \omega v \, dv = \frac{1}{\pi} \int_{-1}^1 \cos \omega v \, dv = \left. \frac{\sin \omega v}{\pi \omega} \right|_{-1}^1$$

$$= \frac{2 \sin \omega v}{\pi \omega}$$

$$B(\omega) = \frac{1}{\pi} \int_{-1}^1 \sin \omega v \, dv = 0.$$

$$\Rightarrow f(x) = \frac{2}{\pi} \int_0^{\infty} \frac{\cos \omega x \sin \omega}{\omega} \, d\omega.$$

Questions of Fourier Series.

$$f(x) = \begin{cases} 0, & -\pi < x < -\pi/2 \\ 1, & -\pi/2 < x < \pi/2 \\ 0, & \pi/2 < x < \pi. \end{cases}$$

let,

$$f(x) = \frac{1}{2} a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx).$$

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx, \quad a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cdot \cos nx \, dx$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cdot \sin nx \, dx.$$

for a_0 ,

$$a_0 = \frac{1}{\pi} \left\{ \int_{-\pi}^{-\pi/2} 0 \, dx + \int_{-\pi/2}^{\pi/2} 1 \, dx + \int_{\pi/2}^{\pi} 0 \, dx \right\}$$

$$= \frac{1}{\pi} \left(0 + \int_{-\pi/2}^{\pi/2} 1 \, dx + 0 \right)$$

$$= \frac{1}{\pi} \left[\frac{x}{1} \right]_{-\pi/2}^{\pi/2} = \frac{2\pi}{2\pi} = 1.$$

for a_n ,

$$a_n = \frac{1}{\pi} \left\{ 0 + \int_{-\pi/2}^{\pi/2} \cos nx \, dx + 0 \right\}$$

$$= \frac{1}{\pi} \left[\frac{\sin nx}{n} \right]_{-\pi/2}^{\pi/2} \Rightarrow \frac{1}{n\pi} \left\{ \sin \frac{n\pi}{2} - \sin \left(-\frac{n\pi}{2} \right) \right\}$$

$$= \frac{1}{n\pi} \cdot \frac{2 \sin n \frac{\pi}{2}}{2} = \frac{2}{n\pi}$$

for b_n ,

$$b_n = \frac{1}{\pi} \left[\int_0^{\pi/2} 0 + \int_{-\pi/2}^{\pi/2} \sin nx \, dx + 0 \right]$$

$$= \frac{1}{n\pi} \left[-\cos nx \right]_{-\pi/2}^{\pi/2}$$

$$= -\frac{1}{n\pi} \left[\frac{\cos n \frac{\pi}{2}}{2} - \frac{\cos n \frac{\pi}{2}}{2} \right]$$

$$= -\frac{1}{n\pi} = 0.$$

Now,

$$f(x) = \frac{1}{2} a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

$$= \frac{1}{2} + \frac{2}{\pi} \cos x + \frac{\cos 2x}{\pi} + \frac{2}{3\pi} \cos 3x + \dots$$

Question.

$$F(x) = \begin{cases} 0 & \text{if } -2 < x < -1 \\ K & \text{if } -1 < x < 1 \\ 0 & \text{if } 1 < x < 2 \end{cases} \quad p = 2L = 4, L = 2$$

Solution.

$$F(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right)$$

$$a_0 = \frac{1}{2L} \int_{-L}^L F(x) dx$$

$$a_n = \frac{1}{L} \int_{-L}^L F(x) \cos \frac{n\pi x}{L} dx$$

$$b_n = \frac{1}{L} \int_{-L}^L F(x) \sin \frac{n\pi x}{L} dx.$$

$$a_0 = \frac{1}{2L} \int_{-L}^L F(x) dx.$$

$$a_0 = \frac{1}{2(2)} \left(\int_{-2}^{-1} 0 dx + \int_{-1}^1 K dx + \int_1^2 0 dx \right)$$

$$a_0 = \frac{1}{4} \left(\int_{-1}^1 K dx \right)$$

$$a_0 = \frac{K}{4} \int_{-1}^1 dx$$

$$a_0 = \frac{K}{4} \left[x \right]_{-1}^1$$

$$a_0 = \frac{K}{4} (1 - (-1))$$

$$a_0 = \frac{K}{4} (2)$$

 $\Rightarrow$

$$a_0 = \frac{K}{2}$$

$$a_n = \frac{1}{L} \int_{-L}^L F(x) \cos \frac{n\pi x}{L} dx$$

$$a_n = \frac{1}{2} \left(\int_{-1}^1 K \cos \frac{n\pi x}{2} dx \right)$$

$$a_n = \frac{K}{2} \left(\int_{-1}^1 \cos \frac{n\pi x}{2} dx \right)$$

$$a_n = \frac{K}{2} \left[\frac{\sin(n\pi x/2)}{n\pi/2} \right]_{-1}^1$$

$$a_n = \frac{K}{2} \left[\sin\left(\frac{n\pi}{2}\right) - \sin\left(-\frac{n\pi}{2}\right) \right] \frac{2}{n\pi}$$

$$a_n = \frac{K}{n\pi} 2 \sin\left(\frac{n\pi}{2}\right) \Rightarrow a_n = \frac{2K}{n\pi} \sin\left(\frac{n\pi}{2}\right)$$

putting values of n in a_n .

$$n=1$$

$$a_1 = \frac{2K}{\pi} \sin\left(\frac{\pi}{2}\right) \Rightarrow a_1 = \frac{2K}{\pi}$$

$$n=2$$

$$a_2 = \frac{2K}{2\pi} \sin\left(\frac{2\pi}{2}\right) \Rightarrow a_2 = \frac{K}{\pi} (0) \Rightarrow a_2 = 0$$

$$n=3$$

$$a_3 = \frac{2K}{3\pi} \sin\left(\frac{3\pi}{2}\right) \Rightarrow a_3 = \frac{2K}{3\pi} (-1) \Rightarrow a_3 = -\frac{2K}{3\pi}$$

$$n=4$$

$$a_4 = \frac{2K}{4\pi} \sin\left(\frac{4\pi}{2}\right) \Rightarrow a_4 = \frac{K}{2\pi} (0) \Rightarrow a_4 = 0$$

$$n=5$$

$$a_5 = \frac{2K}{5\pi} \sin\left(\frac{5\pi}{2}\right) \Rightarrow a_5 = \frac{2K}{5\pi} (1) \Rightarrow a_5 = \frac{2K}{5\pi}$$

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \sin \frac{n\pi x}{L} dx.$$

$$b_n = \frac{1}{2} \int_{-1}^1 K \sin \frac{n\pi x}{2} dx$$

$$b_n = \frac{K}{2} \frac{2}{n\pi} \left[-\cos \frac{n\pi x}{2} \right]_{-1}^1$$

$$b_n = -\frac{K}{n\pi} \left[+\cos \frac{n\pi}{2} - \cos \left(-\frac{n\pi}{2} \right) \right]$$

Now, put the values of n in b_n .

$$n=1$$

$$b_1 = \frac{-K}{\pi} \left[\cos \frac{\pi}{2} - \cos \left(\frac{-\pi}{2} \right) \right] = 0$$

$$n=2$$

$$b_2 = \frac{-K}{2\pi} \left[\cos \left(\frac{2\pi}{2} \right) - \cos \left(\frac{-2\pi}{2} \right) \right] = 0$$

$$n=3$$

$$b_3 = \frac{-K}{3\pi} \left[\cos \left(\frac{3\pi}{2} \right) - \cos \left(\frac{-3\pi}{2} \right) \right] = 0$$

$$n=4$$

$$b_4 = \frac{-K}{4\pi} \left[\cos \left(\frac{4\pi}{2} \right) - \cos \left(\frac{-4\pi}{2} \right) \right] = 0$$

$$n=5$$

$$b_5 = \frac{-K}{5\pi} \left[\cos \left(\frac{5\pi}{2} \right) - \cos \left(\frac{-5\pi}{2} \right) \right] = 0$$

Now, putting all the values in $F(x)$,

$$F(x) = a_0 + \sum_{n=1}^{\infty} \left(\frac{a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L}}{L} \right)$$

$$F(x) = \frac{K}{2} + \frac{2K}{\pi} \cos \frac{\pi x}{2} + \left(\frac{-2K}{3\pi} \right) \cos \frac{3\pi x}{2} + \frac{2K}{5\pi} \cos \frac{5\pi x}{2} + \dots$$

$$F(x) = \frac{K}{2} + \frac{2K}{\pi} \left[\cos \left(\frac{\pi x}{2} \right) - \frac{1}{3} \cos \left(\frac{3\pi x}{2} \right) + \frac{1}{5} \cos \left(\frac{5\pi x}{2} \right) - \dots \right]$$